

WASP-104b

R-2645

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-104_150228 · created 2026-08-29T22:48:08+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2457081.94224 +/- 0.0006 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.151 +/- 0.011
Transit depth (Rp/Rs)^2	2.29 +/- 0.33 [%]
Orbital Inclination (inc)	83.47 +/- 1.31
Airmass coefficient 1 (a1)	1.263 +/- 0.014
Airmass coefficient 2 (a2)	-0.1871 +/- 0.0051
Scatter in the residuals of the lightcurve fit is	1.11 %
Best Comparison Star	None
Optimal Aperture	3.28
Optimal Annulus	9.59
Transit Duration (day)	0.074 +/- 0.012

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.151 ± 0.011 vs 0.1214 (deviation 2.69σ)
Duration fit vs published	0.074 d vs 0.0742 d (deviation 0.01σ)
Mid-time O–C	-1.2 min
Depth SNR	13.7
Residual scatter	1.11 %

Light curve

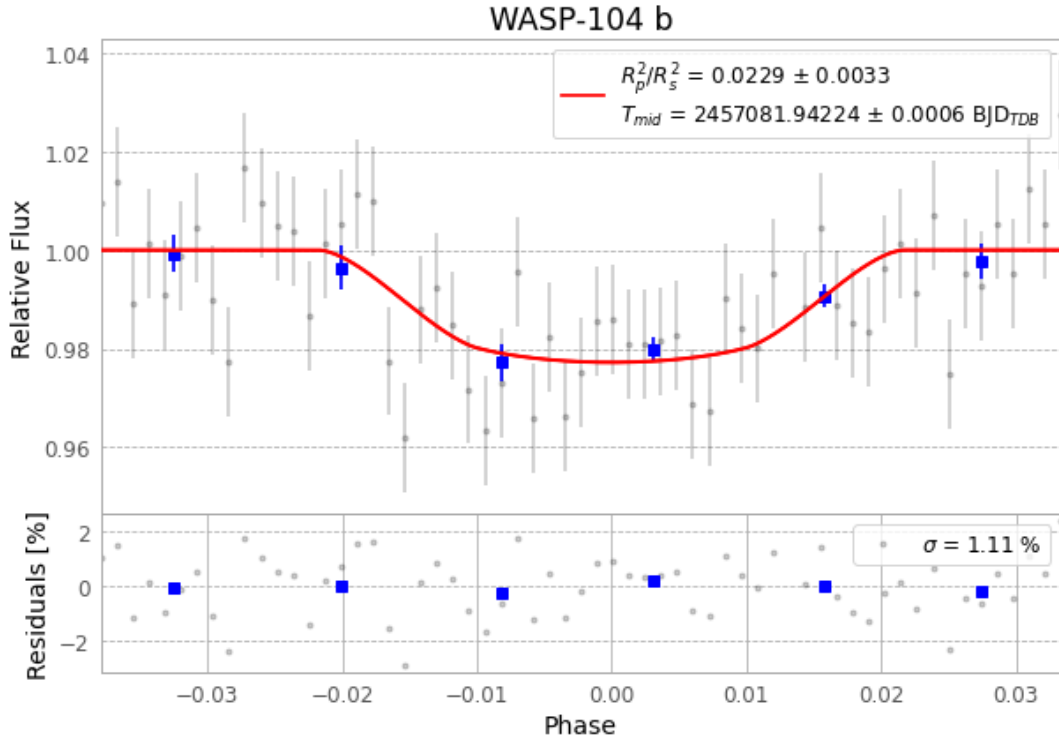


Figure 1 — FinalLightCurve_WASP-104 b_2015-02-28.png (SHA-256 d45dad8c6e60f00f...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [296, 242], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 901fc4b636528e84...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

61 FITS frames (40 MB), WASP-104150228085112.FITS → WASP-104150228115112.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-104-150228.log (191dc2f7fb22...), FinalLightCurve_WASP-104 b_2015-02-28.png (d45dad8c6e60...), FinalLightCurve_WASP-104 b_2015-02-28.csv (db3b4298660e...)

Compute resources

Wall time 133.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-29 22:48Z.